Jinyu Xie

mail Github Google Scholar HomePage

Education

2021 – Present **B.S., UESTC** in Mathematics-Physics Fundamental Science
Yingcai Honors College (Elite program, top 102 students selected from 5000 freshmen).
GPA: 3.87/4.00; Avg. Score: 89.32/100.00.

Research Publications

1. **J.-Y. Xie**, H. Zhang, X. Zhao, and Y. Luo, "IRTF: A new tensor factorization for irregular multidimensional data recovery," *Knowledge-Based Systems*, under review.
2. **J.-Y. Xie***, J.-Y. Li *, Y.-S. Luo, X.-L. Zhao, and J.-L. Wang, "H2TF for hyperspectral image denoising: Where hierarchical nonlinear transform meets hierarchical matrix factorization," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, 2023. DOI: 10.1109/LGRS.2023.3294933.
3. Y.-Y. Liu, X.-L. Zhao, **J.-Y. Xie**, X. Zhi, and G. Vivone, "Bi-level tensor decomposition for hyperspectral image restoration," in *IEEE International Geoscience and Remote Sensing Symposium*, 2024.
4. H. Zhang, T.-Z. Huang, X.-L. Zhao, S. Zhang, **J.-Y. Xie**, T.-X. Jiang, and M. K. Ng, "Learnable transform-assisted tensor decomposition for spatio-irregular multidimensional data recovery," *ACM Transactions on Knowledge Discovery from Data*, forthcoming.

Research Experience

2023/07 – present **Topic: Tensor Decomposition for Irregular Data**. Supervised by Prof. Xile Zhao.
I studied irregular data algorithms and developed a novel tensor decomposition method for irregular data (e.g., spatial transcriptomics data) reconstruction. A paper is submitted based on this work.

2022/07 – present **Topic: Advanced Tensor Decomposition**. Supervised by Prof. Xile Zhao.
I have studied tensor decomposition algorithms for various data modeling and, in collaboration with others, have developed a novel tensor decomposition method that integrates deep learning techniques. Based on this work, I have published a journal article.

2022/01 – 2022/07 **Topic: Deep Learning Algorithms Research**. Supervised by Assoc. Prof. Liangjian Deng.
I studied convolutional neural network (CNN) algorithms for super-resolution and deraining and collaborated on an orange quality classifier using CNNs. I then learned spiking neural network (SNN) theories and worked on implementing SNNs with memristors and mathematical modeling of the combination.

Patents

2024/04 H. Zhang, T.-Z. Huang, X.-L. Zhao, S. Zhang, J.-Y. Xie, T.-X. Jiang, and M. K. Ng. "Irregular high-dimensional data restoration method and system based on transform domain tensor decomposition". CN117911802A, filed January 17, 2024, and issued April 19, 2024.

Awards

2023/03 Second Prize, 14th Mathematics Competition of Chinese College Students (Mathematics Class A).
2023/12 **Excellent**, University Student Innovation and Entrepreneurship Competition (continuous funding).
2023/05 **First Prize**, 23rd Mathematical Modeling Contest, UESTC.
2022/12 **Outstanding**, 2021 Freshman Extracurricular Innovation Project, UESTC.
2023/06 **Outstanding**, University Student Innovation and Entrepreneurship Training Program, UESTC.
2024/6 **Exemplary Student Scholarship**, Exemplary Student Scholarship, UESTC (three times).

Skills

Coding Python, Matlab, LaTeX, Markdown.

电子科技大学本科生成绩单

Official Undergraduate Transcript of University of Electronic Science and Technology of China

N0.2006,Xiyuan Ave,
West Hi-tech Zone
Chengdu,Sichuan 611731
P.R.China

Name:Xie Jinyu | Student ID:2021010911011 | Gender:Male

Date of Birth:2002-12-20 | Date of Enrollment:2021-08-26 | Duration:4 Years

School:Yingcai Honors College | Major:Mathematics-Physics Fundamental Science (Yingcai Honors Program of UESTC)

Compulsory Courses

Courses	Term	Credit	Score	GP	Courses	Term	Credit	Score	GP
Programming Design Methods	2021-2022-1	3	62	1.7	Advanced Algebra and Geometry I	2021-2022-1	4	96	4.0
Advanced Calculus I	2021-2022-1	5	97	4.0	Military Theory	2021-2022-1	2	89	4.0
Military Training	2021-2022-1	2	P	4.0	Table tennis A	2021-2022-1	1	99	4.0
Introduction to Artificial Intelligence	2021-2022-1	1	75	3.0	Appreciation of Human Civilization	2021-2022-1	1	B	4.0
Ideology and morality and the rule of law	2021-2022-1	3	96	4.0	General English	2021-2022-1	3	71	2.6
mental health and the ability of innovation	2021-2022-1	2	93	4.0	ACM-ICPC Algorithm and Program Design	2021-2022-2	2	92	4.0
University Physics I	2021-2022-2	4	94	4.0	Physical Experiment I	2021-2022-2	2	90	4.0
Special Lecture on the Historiography in Reform and Opening up	2021-2022-2	1	83	3.8	Advanced Algebra and Geometry II	2021-2022-2	4	92	4.0
Advanced Calculus II	2021-2022-2	5	96	4.0	Mathematics and creativity(H)	2021-2022-2	2	94	4.0
Exploring Information Mystery（PBLF）	2021-2022-2	1	99	4.0	Information Science & Technology and World Progress	2021-2022-2	2	88	4.0
The Summary of Chinese Modern History	2021-2022-2	3	97	4.0	footballB	2021-2022-2	1	84	3.9
University Physics II	2022-2023-1	4	93	4.0	Physical Experiment II	2022-2023-1	2	86	4.0
Functions of Complex Variables	2022-2023-1	3	90	4.0	Advanced Calculus III	2022-2023-1	4	96	4.0
Discrete Mathematics	2022-2023-1	3	88	4.0	Theoretical Mechanics	2022-2023-1	3	86	4.0
Table Tennis C	2022-2023-1	1	95	4.0	Data Structure and Algorithm	2022-2023-1	3	86	4.0
Physical integrative innovation project(H)	2022-2023-1	1	86	4.0	English For Academic Reading	2022-2023-1	2	83	3.8
Algorithms for Optimization	2022-2023-1	3	92	4.0	Drama Appreciation	2022-2023-2	2	86	4.0
Probability and Statistics	2022-2023-2	5	88	4.0	Basic Academic Training I	2022-2023-2	2	100	4.0
Table Tennis D	2022-2023-2	1	90	4.0	Artificial Intelligence	2022-2023-2	3	88	4.0
Engineering practice innovation project(H)	2022-2023-2	2	85	4.0	Mathematical Modeling	2022-2023-2	3	79	3.4

Office of Academic Affairs
UESTC

Print Date:2024-05-20

University of Electronic Science and Technology of China 电子科技大学教务处 学生成绩证明专用章 Office of Academic Affairs Achievements Certification

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Courses	Term	Credit	Score	GP	Courses	Term	Credit	Score	GP
Numerical Analysis	2022-2023-2	3	79	3.4	Elements of Information Theory	2022-2023-2	2	87	4.0
English For Academic Writing and Communication	2022-2023-2	2	85	4.0	Marketing Management:Internet Thinking and its Application	2022-2023-2	2	90	4.0
Multivariate Statistical Analysis	2023-2024-1	3	93	4.0	Basic principles of Marxism	2023-2024-1	3	80	3.5
Comprehensive Experiment of Artifical Intelligence	2023-2024-1	1	85	4.0	Time Series Analysis	2023-2024-1	3	88	4.0
Digital Image Processing	2023-2024-1	3	91	4.0					
Elective Courses									
From Nobel Prize to Multi-Billion Dollar Industry: A Brief History of the Information Age	2021-2022-2	1	93	4.0	Award of the Second Rank in Province/Ministry-level Competitions	2022-2023-2	2	91	4.0
Get a paper published in an International high-level journal or international top academic conference	2023-2024-1	6	95	4.0	Graphing techniques in scientific research	2023-2024-1	2.5	93	4.0
Stress and Emotion Management	2023-2024-2	1	87	4.0					

Weighted Average: 89.32
GPA: 3.87

Note: △ represents make-up score, ※ represents re-study score

Office of Academic Affairs
UESTC

Print Date:2024-05-20

学生成绩证明专用章
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Achievements Certification